

What Can Mongolian Tell Us about Binding?¹

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Abstract

In Mongolian, division of labor is clear for the anaphor *өөр* and the reflexive-possessive suffix, *-aa*, the latter being the hallmark of reflexive binding: *өөр* marks binding not intertwined with verbal inflections and *-aa* marks it when it is intertwined with verbal inflections. The morpho-syntactic properties of Mongolian binding strategies centered on this property can tell us the following. **First**, binding is dynamic and derivational process; it is an Agree relation between phi-features; Agree may be followed by Move of the binder from the predicate-internal position (Spec of VoiceP) to a position outside the predicate (Spec of TP); when not followed by Move, Agree makes up predicate-internal binding; when Agree plus Move takes place, predicate-external binding is implemented. **Second**, TP as the binding domain as identified in the pre-minimalist binding theory (the Standard Binding Theory (Chomsky 1981)) holds true for predicate-external binding, while the minimal phase boundary as a binding domain as in the late minimalist era holds for predicate-internal binding. **Third**, The Anaphor Agreement Effect follows from the “internal-hood” of the binder, which is “immature” as a subject, with the inflectional properties of the predicate is yet to be assigned, but not from a capsule-like structure of anaphors. **Fourth**, anaphors are possessive DPs structurally, which are formed in a portmanteau fashion, where phi-features that sit in the specifier of the possessive D head are valued by those on antecedent nominals occupying Spec of a predicate phrase (VoiceP). **Fifth**, reflexivity, which is conditioned by binding as feature-valuation (Agree), is thus a Spec-Spec (subject-subject or subject-possessor) relation, rather than a Spec-Comp (subject-object) relation.

Keywords predicate-internal/external binding, reflexive, possessive, anaphor, Agree

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1 Introduction

This paper explicates the nature of reflexive binding in Mongolian and demonstrates its theoretical implications for our understanding of binding in general.

In syntax, the shift from the pre-minimalist (i.e., Government & Binding/GB) to the minimalist (i.e., Minimalist Program/MP) era saw binding theory move from rule-bound modules (like Binding Principles A, B, C) to a combination of economy principles and feature-checking under minimalist operations (e.g., Agree and Move/Merge). Pre-minimalism regarded binding domains to be defined by a Governing Category, which involves the anaphor, its governor, and a subject, whereas in (later) minimalist syntax, a phase, which is a cyclic unit in syntactic derivation, provides a way to define binding domains (Lee-Schoenfeld 2004, 2008; Canac-Marquis 2005; Hicks 2009; Gallego 2010; Safir 2011; Antonenko 2012; Bošković 2016). Thus, the binding domain for the anaphor in sentences such as (1) underwent a shift from TP, the inflectional phrase, to vP, the thematic phrase.

(1) Pasha loves herself: [TP Pasha_i ... herself_i] → [vP=_{phase} Pasha_i ... herself_i]

While the minimalist explanation of binding is superior to the pre-minimalist approach by virtue of taking it as streamlined within the derivation of the clause structure, controversies, in essence, lie in what mechanism it is exactly and how it is implemented. At least three distinct approaches from the minimalist literature on binding can be mentioned: the feature-valuation approach (Rauand 2011), the feature-transmission approach (Kratzer 2009), and the feature-matching approach (Preminger 2019). The feature-valuation approach, also called the reductionist approach, views binding as reducible to Agree, at least for SE-anaphors, where they originate as elements underspecified for phi-features, which are unvalued but matched and valued by phi-features of the binder via an Agree relation. The feature-transmission approach holds that the binder transmits its phi-features to the bound anaphor, which enters the derivation as a featureless minimal pronoun. The feature-matching approach, also called the encapsulation approach, arguing against the reductionist approach, holds that full agreement in phi-features between the anaphor and its antecedent is prevented by a structural layer (AnaphP) on top of the phi-feature layer, which is thus encapsulated, and that binding is feature-matching, not feature-valuation, within opaque domains.

Another controversial issue is about which of vP and TP is the right site in which to implement binding. Kratzer (2009), Rooryck and Vanden Wyngaerd (2011), and others argue that phi-features of anaphors start out from the specifier of vP, which is thus the binding domain. For Reuland (2011), however, binding is intertwined with Case of a local subject and implemented through Agree within vP and Move within TP. That is, anaphors find their antecedents within vP, which, being a phase, serves as the structural boundary for Agree, but binding as Agree is completed within TP, where a linked chain is formed between anaphors, being phi-deficient, and their antecedents, being structural-Case-marked and phi-complete (Reuland and Everaert 2000; Reuland 2001, 2011, 2020a, 2020b). In this sense, TP could still be regarded as a domain for binding, echoing the pre-minimalist treatment of binding domain, although Reuland attempts to determine boundaries for binding by Agree and chain formation and avoids explicitly identifying an inflected phrase such as TP as a binding domain.

There also exist controversies about the nature of anaphors. For Kratzer (2009), anaphors enter the derivation as “minimal pronouns” lacking inherent phi-features and get phi-values by being bound by λ -operators on functional heads, which acts as a conduit for feature-

transmission, while for Reuland (2011), they are born with unvalued phi-features, which may be deficient, and get phi-values by being involved in chains formed between anaphors and their antecedents. For Preminger (2019), anaphors are capsule-like structures, where the phi-layer is encapsulated by the anaphor-layer on top of it and therefore fails to fully agree with a nominal in a higher position.

As will be clear, the first and the third controversies arise where anaphors are not correctly analyzed as possessive DPs formed in a portmanteau fashion, and the second arises where predicate-internal binding is undistinguished from predicate-external binding. None of the above-mentioned studies have treated anaphors as possessive DPs and none have explicitly identified binding as predicate-internal or predicate-external.

Importantly, while, in the pre-minimalist era, verbal inflection was often taken as the hallmark of a boundary for syntactic binding, with inflected clauses generally allowing less freedom for anaphors than uninflected clauses, the role of inflection has been unclear in resolving the above-mentioned controversial issues in the minimalist era. In Reuland's approach, verbal inflection plays a role in forming a feature-chain, but it is not a factor that can affect binding in other approaches. The connection between verbal inflection and binding has been removed or at least loosened by the combination of binding with the notion of phase in minimalist syntax.

This paper shows that verbal inflection and a phase are equally crucial in implementing binding as a derivational process, not a representational constraint. It will be shown that in syntax there are two types of binding, predicate-external and predicate-internal, and that verbal inflection is a key factor in deriving the former from the latter. Mongolian data will tell us that the distinction between the two types of binding can be morphologically realized where possible.

This paper is organized as follows. **Sections 2 and 3** overview the basic properties of the anaphor *öör* and the reflexive-possessive suffix *-aa* in Mongolian, focusing on the conditions and/or constraints on their positions and interactions. **Section 2** illustrates that the anaphor *öör* functions as an object, a possessor, and an embedded subject and that it can be absent at PF when RX is present elsewhere as a reflexive marker. **Section 3** explains that RX is required by conjoint reference but blocked by disjoint reference and that it is licensed by the subject of a local inflected clause. **Section 4** justifies that reflexivity arises between specifiers, not between a specifier and a complement. It will be demonstrated that anaphors are possessive DPs structurally, where the specifier of DP accommodates the element that is subject to binding by a subject from a specifier high up in the predicate phrase. This leads to the structural resemblance between anaphoric objects and typical possessive DPs containing the anaphor. **Section 5** presents a “portmanteau” approach to the formation of possessive pronouns and anaphors, arguing that they are formed through clustering of relevant features including [phi], [possessive], and [reflexive] on independent D heads. D heads undergo fusion to become a single head, followed by vocabulary insertion, thereby forming an anaphor. **Section 6** presents argumentation that for a binder which is an in-situ subject within the predicate, binding is just Agree in phi-features between it and a c-commanded anaphor, hence the label “predicate-internal binding”, while for a binder which is a derived subject outside the predicate, binding is Agree plus Move, hence the label “predicate-external binding”. It will be shown that reflexive binding always holds between two Specs, of which the higher one is Spec of vP, occupied by a subject, and the lower one is either Spec of DP, occupied by a possessor or a genitive embedded

subject, or Spec of CP, occupied by an accusative embedded subject. **Section 7** concludes the paper.

2 Basics of the anaphor *öör*

Öör serves primarily three grammatical roles: object, possessor and subject.² We discuss each situation in sections 2.1, 2.2 and 2.3, and summarize them in section 2.4.

2.1. *Öör* as an object

When used as an object, *öör* may refer to a local subject, where the reflexive-possessive suffix *-aa*, notated as RX,³ is present, as in (2) and (3), or a long-distance subject, where the (non-reflexive) possessive suffix *-ni*, notated as PSS, is present, as in (4).

- (2) Dorž *öör-ig-öö* *šüümjil-sen*.⁴
 Dorž-NOM self-ACC-**RX** criticize-PST
 ‘Dorž criticized himself.’
- (3) Baatar_i Dorž_j-in *öör*_{i/j}-ig-öö* *šüümjil-sen-ig* har-san.
 Baatar-NOM Dorž-GEN self-ACC-**RX** criticize-PST-ACC see-PST
 ‘Baatar saw that Dorž criticized himself (= Dorž).’
- (4) Baatar_i Dorž_j-in *öör_i/*_j-ig-ni* *šüümjil-sen-ig* har-san.
 Baatar-NOM Dorž-GEN self-ACC-PSS[3] criticize-PST-ACC see-PST
 ‘Baatar saw that Dorž criticized him (= Baatar).’

2.2. *Öör* as a possessor

When used as a possessor, *öör*, taking genitive case, may be absent on the surface. Importantly, RX must be present on the possessed noun, as in (5). However, when present, the genitive *öör-in* can host RX, as in (6).

- (5) Baatar (*öör-in*) nom-*ø-oo*⁵ mart-san.
 Baatar-NOM self-GEN book-ACC-**RX** forget-PST
 ‘Baatar forgot his own book.’
- (6) Baatar *öör-in-öö*⁶ nom-ig mart-san.
 Baatar-NOM self-GEN-**RX** book-ACC forget-PST
 ‘Baatar forgot his own book.’

² *Öör*, being a singular form, is specified for number but not for person and gender. The plural form is *öör-sd* (or *öör-üüd*, which is used interchangeably in colloquial Mongolian in eastern Inner Mongolia), which takes a plural antecedent but is not specified for person and gender. The grammatical functions of the singular and plural forms are entirely the same; both can function as objects, possessors, and subjects.

³ **RX**, being subject to vowel harmony, has three allophonic morphemes, *-aa*, *-ee*, *-oo* and *-öö*, which do not differ from each other syntactically and semantically.

⁴ The abbreviations used in this paper include ACC: accusative, ASP: aspect; CS: causative; DAT: dative, GEN: genitive, INS: instrumental; NFIN: non-finite, NOM: nominative, PS: passive, PST: past, and RX: reflexive-possessive suffix.

⁵ When followed by RX, the accusative marker *-ig* is not pronounced except when it is attached to the anaphor *öör*. We use “-ø” to indicate its lack of phonological content.

⁶ For phonological reasons, often the consonant *h* is often inserted between the *n* and a vowel, as in *öör-in-h-öö*.

2.3. *Öör* as the subject of subordinate clauses

Both object clauses, with or without the complementizer *gež*, and object relative clauses can have *öör* as their subject and RX must be present. Situations differ according to the types of clause.

When used as the subject of object clauses headed by *gež*, *öör* takes accusative case and cannot be absent (7). Without *öör*, the covert embedded subject is interpreted as a nonspecific person.

- (7) Baatar *öör-ig-öö* buruud-san *gež* med-sen. (object clause)
 Baatar-NOM self-ACC-RX go wrong-PST COMP know-PST
 ‘Baatar realized that he was wrong.’

When used as the subject of object clauses headed by *gež* and relative clauses, *öör* takes genitive case and can be absent.⁷ RX is present either on *öör*, as in (8) and (10), when it is present, or on the verb, as in (9) and (11).

- (8) Baatar *öör-in-öö* buruud-san-ig meder-sen. (object clause)
 Baatar-NOM self-GEN-RX go wrong-PST-ACC admit-PST
 ‘Baatar admitted that he was wrong.’
- (9) Baatar (*öör-in*) buruud-san-aa meder-sen. (object clause)
 Baatar-NOM self-GEN go wrong-PST-RX admit-PST
 ‘Baatar admitted that he was wrong.’
- (10) Baatar *öör-in-öö* hel-sen *üg-ig* mart-san. (relative clause)
 Baatar-NOM self-GEN-RX say-PST word-ACC forget-PST
 ‘Baatar forgot the words he said.’
- (11) Baatar (*öör-in*) hel-sen *üg-ø-ee* mart-san. (relative clause)
 Baatar-NOM self-GEN say-PST word-ACC-RX forget-PST
 ‘Baatar forgot the words he said.’

2.4. Interim summary

In summary, as shown in (41), *öör-in* (genitive) acts as a possessor in DPs and as the subject of object clauses without *gež* and relative clauses, which are categorially nominalized TP (see section 6). *Öör-ig* (accusative) acts as an object and as the subject of object clauses with *gež*, which are categorially CP (see section 6).

(12) Distributional paradigm of *öör* (interaction of its case and grammatical functions)

	As an object	As a possessor	As a subject
Genitive		possessive DP	object clause without <i>gež</i> ; relative clause
Accusative	DP object		object clause with <i>gež</i>

⁷ The presence or absence of *öör* does not affect the referentiality-related interpretation of the sentence. Its presence may bring an intensifying effect on the possessor. My survey utilizing Inner Mongolia University’s corpus *Juun tümen üg-tei odo üye-in Mongol helen-nii temdeg hadasan material-in hömrög* (Tagged corpus of modern Mongolian, with one million words) shows that in definitely most cases, the anaphor is absent, with RX occurring independently. I searched for four frequently used nouns, *aab* ‘father’, *eež* ‘mother’, *ger* ‘house’, and *mori* ‘horse’, which are RX-marked, in the corpus and found 311 clauses in total but only five out of 311 feature the cooccurrence of *öör-in* and RX.

Regarding the position of RX, which is required for an overt *öör*, two positions are available. One follows the genitive and accusative case markers and the other follows the whole phrase.

(13) Presence or absence of *öör-in* (genitive) and position of RX

Genitive		Accusative
<i>öör-in</i> -RX DP	(<i>öör-in</i>) DP-RX	<i>öör-ig</i> -RX
absence disallowed	absence allowed	absence disallowed

Importantly, the genitive *öör-in* is absent at PF when RX occurs in the second type of position. In other words, *öör-in* is absent when it is separated from RX. This means that RX always requires an overt host. Let us tentatively formulate this property as follows, without theoretical implications at the moment.

(14) The Elsewhere Marker Condition (EMC)⁸

Anaphors can be absent at PF when RX is present elsewhere as a reflexive marker.

3 Basics of the reflexive-possessive suffix *-aa*

As seen in section 2, *öör* always calls for the presence of RX, which plays the most crucial role in reflexive binding in Mongolian. Only elements other than nominative subjects can be attached by RX. This paper, however, is concerned only with cases in which RX is attached to accusative-marked objects.

3.1. Co/Disjoint reference and RX

This section focuses on the interactions of RX and the conjoint reference (CR) or disjoint reference (DR) between subjects or possessors. As can be summarized from section 2, any of the following conditions must be satisfied where RX occurs.

(15) Conditions on RX

- The accusative *öör* acts as an object that is coreferential with a local subject.
- The genitive *öör* acts as the possessor of an NP hosting RX and is coreferential with a local subject.
- The genitive *öör* acts as the subject of a relative clause that modifies an NP hosting RX and is coreferential with the matrix subject.
- The genitive *öör* acts as the subject of a complement clause that contains with its predicate hosting RX and is coreferential with the matrix subject.

Notably, the necessary condition of licensing RX is that coreferentiality is obtained between a subject and *öör*, either accusative or genitive and either present or absent at PF. For (15c), (15d), and (15e), *öör* itself can host RX, which, however, is subject to pragmatic semantic constraints to some extent.

(16) Distributional paradigm of RX in objects (simplex anaphors) and object clauses with *gež*⁹

NOM	ACC	Example
SBJ _i	<i>öör</i> _i -RX	(2, 7)

(17) Distributional paradigm of RX in possessive NPs

⁸ EMC is evidently correct in terms of the linear structure. It is, however, not simply a surface condition for a linear structure. The logic underlying it is essentially a matter of syntax and is hierarchically derived. Detailed elaboration of this issue is left out in this paper.

⁹ Case suffixes are all left out in these tables.

NOM	GEN	ACC	Example
NP2 _i	<i>öör_i</i> -RX	NP _j	(6)
NP2 _i	<i>öör_i</i>	NP _j -RX	(5)
NP2 _i		NP _j -RX	(5)

(18) Distributional paradigm of RX in NPs with relative clauses

NOM	GEN	ACC	Example
NP2 _i	<i>öör_i</i> -RX	NP _{RELj}	(10)
NP2 _i	<i>öör_i</i>	NP _{RELj} -RX	(11)
NP2 _i		NP _{RELj} -RX	(11)

(19) Distributional paradigm of RX in object clauses without *gež*

NOM	GEN	ACC	Example
SBJ _i	<i>öör_i</i> -RX	V	(8)
SBJ _i	<i>öör_i</i>	V-RX	(9)
SBJ _i		V-RX	(9)

It then follows that CR feeds RX and DR blocks it. Importantly, RX is licensed by a local subject that is involved in coreferentiality. In what follows, we test this result.

In (20), *muur* ‘cat’ is possessed by the embedded genitive subject *Bat*, which is in the same clause as *muur*. RX is thus licensed by *Bat* and present on *muur*. Note that this RX is not licensed by the matrix subject *Baatar* because *muur* ‘cat’, the host of RX, is not an argument of the matrix verb; that is, the host of RX and the matrix subject are not coarguments and are not in the same clause. Therefore, DR between the matrix subject and the embedded subject or between the matrix subject and the host of RX is not relevant here and does not block RX.

- (20) Baatar Bat-in muur-*φ*-aa üns-sen-ig har-san.
Baatar-NOM Bat-GEN cat-ACC-**RX** kiss-PST-ACC see-PST
‘Baatar saw that Bat kissed his cat (= Bat’s cat).’

In (21), RX on *muur* is licensed by the embedded subject *bagš* ‘teacher’, not by the matrix subject *Baatar*. In contrast, RX on the embedded subject *bagš* is licensed by the matrix subject because there is a possessive relation between them.

- (21) Baatar bagš-in-aa muur-*φ*-aa üns-sen-ig har-san.
Baatar-NOM teacher-GEN-**RX** cat-ACC-**RX** kiss-PST-ACC see-PST
‘Baatar saw that his teacher (= Baatar’s teacher) kissed his cat (= teacher’s cat).’

The same holds true in sentences containing a relative clause, as in (22).

- (22) Baatar Bat-in bagš-aas-aa zeel-sen nom-ig-ni unš-sen.
Baatar-NOM Bat-GEN teacher-ABL-**RX** borrow-PST book-ACC-PSS[3] read-PST
‘Baatar read the book that Bat borrowed from his teacher (= Bat’s teacher).’

In conclusion, RX is licensed by the subject of a local inflected clause but blocked by DR.

3.2. Licensing domain for RX

As seen in section 3.1, RX is licensed by a local subject. This subsection elaborates on this. As exemplified in (23), RX on *öör* is licensed by the embedded subject *Dorž*, which is coreferential with *öör*.

- (23) Baatar_i Dorž_j-ig öör_{i/j}-ig-**öö** šüümjil-sen-ig har-san.
Baatar-NOM Dorž-ACC self-ACC-**RX** criticize-PST-ACC see-PST

‘Baatar saw that Dorž criticized himself (= Dorž).’

If *öör* does not refer to the embedded subject *Dorž*, but the matrix subject *Baatar*, RX is not present on *öör*, but PSS is, as in (24).

- (24) Baatar_i Dorž_j-ig öör_{i/*j}-ig-ni šüümjil-sen-ig med-ne.
 Baatar-NOM Dorž-ACC self-ACC-PSS[3] criticize-PST-ACC know-NPST
 ‘Baatar knows that Dorž criticized him (= Baatar).’

The same holds true of the case in which the embedded clause is headed by the complementizer *gež*, as exemplified in (25) and (26).

- (25) Baatar_i Dorž_j-ig öör_{i/*j}-ig-öö šüümjil-sen gež hel-sen.
 Baatar-NOM Dorž-ACC self-ACC-RX criticize-PST COMP say-PST
 ‘Baatar said that Dorž criticized himself (= Dorž).’
 (26) Baatar_i Dorž_j-ig öör_{i/*j}-ig-ni šüümjil-sen gež hel-sen.
 Baatar-NOM Dorž-ACC self-ACC-PSS[3] criticize-PST COMP say-PST
 ‘Baatar said that Dorž criticized him (= Baatar).’

Note that where RX is blocked, *-ni*, which is a marker of possessive (not reflexive or reflexive-possessive) relation, is needed. *-Ni* is not licensed by a subject of any type and is not subject to clause boundaries. Note that *öör* with *-ni* is coreferential with the matrix subject *Baatar*. However, this does not mean that the matrix subject licenses *-ni*. This is because the host of *-ni* can also be possessed by an element other than the matrix subject and is not obligatory, as shown by the following example, in which *-ni* is glossed by “PSS[3]” (for “third person possessive marker”).¹⁰

- (27) Baatar_i Dorž-in hüü-ig(-ni) šüümjil-sen.
 Baatar-NOM Dorž-GEN son-ACC-PSS[3] criticize-PST
 ‘Baatar criticized Dorž’s son.’

Importantly, RX is present only when there is verbal inflection; when the subject of an uninflected predicate binds the anaphor, PSS rather than RX is present, as evidenced by the following examples, in which the agent *Dorž*, which is the subject of uninflected predicates, cannot license RX.

- (28) Baatar_i Dorž_j-oor (öör_{i/*j}-in) nom- ϕ -oo unš-uul-san.
 Baatar-NOM Dorž-INS self-GEN book-ACC-RX read-CS-PST
 ‘Baatar made Dorž read his book (=Baatar’s book).’
 (29) Baatar_i Dorž_j-oor (öör_{i/*j}-in) nom- ϕ -ni unš-uul-san.
 Baatar-NOM Dorž-INS self-GEN book-ACC-PSS[3] read-CS-PST
 ‘Baatar made Dorž read his book (=Dorž’s book).’

All this means that the licensing domain for RX is a minimal inflected clause. Categorially, it is TP. The subject of TP, however, is not necessarily nominative in Mongolian. Rather, it is primarily accusative or genitive, as in (7–11) and (20–29).

4. Reflexivity as a Spec-Spec relation

As seen in section 2, the anaphor may be either a subject, or a possessor, or an object and there are three kinds of coreferential relations: subject-subject relation (SSR), subject-possessor

¹⁰ Unlike RX, PSS[3] is attached to a third person element. Mongolian also has PSS[1] and PSS[2]. For relevant descriptions, see Janhunen (2012) and Kullmann and Tserenpil (2015) among many.

relation (SPR) and subject-object relation (SOR). What is the common property of SSR, SPR and SOR? What is the most basic relation that underlies SSR, SPR and SOR, whereby they are expressed by the common marker RX? This issue naturally relates to what reflexivity is.

In line with extensive discussion in the literature (Faltz 1977; Reuland and Reinhart 1993; Kazenin 2001; Huang 2005; Reuland 2011, and many), an informal characterization of reflexivity would be like (30) below.

(30) Reflexivity is a dependency between two arguments of a predicate that assigns external and internal roles to them.

One of the most influential studies on reflexivity is Reinhart and Reuland (1993) as well as their subsequent research. Reuland and Reinhart (henceforth, R&R) defined reflexivity as follows.

(31) A predicate is reflexive iff two of its arguments are bound by the same λ -operator. (Reuland 2014: 11)

Roughly speaking, reflexivity is obtained between a subject and an object, according to (30) and (31). Accordingly, only SOR among the three relations noted above would qualify as reflexivity.

However, as suggested by the Mongolian data discussed in section 3, reflexivity is not necessarily SOR. Reflexivity as defined by R&R therefore remains one in the narrowest sense. Reflexivity identified in a more general sense should be obtained between a subject and a Spec element in the object phrase. This is indicated by the high productivity of body-anaphors in many languages including Mongolian.

Note that bodypart anaphors such as *öör-in bey* ‘self’s body’ in Mongolian are possessive DPs structurally. In a sense, *bey*, as the lexical core of the anaphor, is possessed by *öör*, which carries the referential meaning of the anaphor, as illustrated below.

(32) Baatar_i *öör*_{i/*j}-in *bey*- ϕ -ee *šüümjil*-sen.
 Baatar-NOM self-GEN body-ACC-RX criticize-PST
 ‘Baatar criticized himself.’

In this sense, structurally, *bey* is N, selected by *öör-in*, which is a possessive determiner D, as represented below.

(33) [XP ... subject ... [DP possessor [D' D [N]]]

(34) [XP ... Baatar ... [DP *öör* [D' -in [*bey*]]]

This suggests that the bare anaphors *öör* and *bey*, in (35) and (36) are in fact also realizations of the possessive structure in (33). For *öör*, D and N have no content at PF, and for *bey*, Spec of DP and D have no content.

(35) Baatar_i *öör*_{i/*j}-ig-*öö* *šüümjil*-sen.
 Baatar-NOM self-ACC-RX criticize-PST
 ‘Baatar criticized himself.’

(36) Baatar_i *bey*_{i/*j}- ϕ -ee *šüümjil*-sen.
 Baatar-NOM body-ACC-RX criticize-PST
 ‘Baatar criticized himself.’

Cross-linguistically, possessive-structured body-anaphors are readily available, for example, *kò a li* ‘his body’ in Haitian Creole (Rooryck and Vanden Wyngaerd 2011: 40), *zi-shen* ‘self’s body’ in Mandarin, and *zi-sin* ‘self’s body’ in Japanese.¹¹ These anaphors all behave in a very similar

¹¹ See Bai (2024, manuscript) for detailed discussion on Chinese and Japanese anaphors.

way to Mongolian *öör-in bey*. They all literally mean “self’s body”. Differences among such languages in the morpho-syntactic properties of anaphors are supposedly related to their lexicalization, which took place diachronically, as well as language-specific ways of deriving pronouns and anaphors. Under a framework such as Distributed Morphology (Halle and Marantz 1993; Embick and Marantz 2008), all these elements can be analyzed as spelling out one or more terminal nodes resulting from an operation such as fusion.

Conceptually, a possessive-possessum relation is a kind of subject-object relation mediated by a functor. In this regard, the predicate is something like “ x holds y ”, where *holds* is a possessive predicate, x is an external argument,¹² and y is an internal argument. Roversi (2024) observes that an abstract transitive verb, what he calls POSS, takes the possessor as its external argument and the possessum as its internal argument, which gives rise to the same possessive semantics as the nominal head *ʒ*.

(37) [_{VP} possessor [_V POSS [_{DP} possessum]]] (Roversi 2024: 2)

Roversi (2024) argues that in languages such as Äiwoo possessive DPs are derived by relativization of the possessum. Let us illustrate this using English examples.

(38) Derivation of *my book*:

[_{DP} book_i [_{RC} ... [_{VP} I [_V POSS [_{DP} ____ _i]]]]] (based on Roversi 2024)

This derivation, according to Roversi (2024), is not supposed to be universal. However, it provides us with a way into the logical structure of possessums. Based on (38), we paraphrase (39) as (40) with the relevant structure represented by (41).

(39) Pasha loves her cat.

(40) Pasha loves the cat she holds/possesses/has.

(41) [_{XP} Pasha_j [_{VP} love [_{DP} cat_i [_{RC} [_{VP} she_j [_V ____ _i]]]]]]]

Note that we are not trying to syntactically derive possessive DPs by relativization. We are representing their logically interpreted structure instead. Given this, reflexivity arises between a subject and a possessor, which is the subject of the semantic predicate *HOLD* (or *POSS*), but not between a subject and a possessum. Specifically, the following is obtained.

(42) Reflexivity arises between specifiers, not between a specifier and a complement. (to be revised)

This brings about SSR, SPR and SOR under a unified frame, which are distinct instances of the Spec-H-Comp structure, as illustrated below.

(43) [_{XP} Spec ... [_{HP} Spec [_H H [Comp]]] ...] (for all)

a. [_{VP} sbj_i ... [_{VP} sbj_i [_V V [obj]]] ...] (for SSR)

b. [_{VP} sbj_i ... [_{DP} possessor_i [_D D^[HOLD] [N]]] ...] (for SPR and SOR)

Thus, both possessive predicates as in (44) and anaphoric predicates as in (45) can be successfully captured by (46a) in a unified way.

(44) Pasha_i loves her_i cat.

¹² R&R (1993) and many others also treat the possessor in a possessive DP as an external argument/subject. By this, R&R (1993: 682) explained the difference in grammaticality between the following sentences. In (ib), *your* is present as the subject of a semantic predicate and therefore subject to their Condition B, leading to the ungrammaticality, with no coreferentiality obtained between it and *herself*.

(i) a. Lucie liked [(a) picture of herself].

b. */?Lucie liked [your picture of herself].

(46) A unified structure of anaphoric and possessive reflexives (informal):

- The following is an elaboration on the structure of anaphors.

- D (=’s) is a possessive determiner head and a semantic predicate *HOLD*;
- The referential part, that is, phi-features, occupy Spec of DP;
- [phi] on Spec and [poss] on D (probably plus a Case feature) are spelled out as *her*;
- Two predicates (one being matrix and syntactic, another being embedded and semantic) are involved in the sentence;
- Neither of the two predicate is reflexive;
- Reflexivity obtains between the matrix subject *Pasha* and the logical subject *SHE*, which is contained by the DP and therefore is assigned no semantic role by the matrix predicate and is not bound by its λ -operator.

(48) Baatar öör-in bey-*φ*-ee šüümjil-sen.
Baatar-NOM self-GEN body-ACC-**RX** criticize-PST
'Baatar criticized himself.'
(= (32))

(50) A unified structure of anaphoric and possessive reflexives (informal):

- Combining this with the fact that subjects are Specs and embedded subjects can be coreferential with matrix subjects (sections 2.3 and 3), we can conclude that reflexivity is a Spec-Spec relation, not a Spec-Comp relation.¹⁴ Specifically, the phi-features of a subject, which occupies the Spec position and those occupying the Spec of an anaphoric phrase (DP) come to match, as will be seen in section 6, thereby yielding reflexivity. Combining this with the conjecture that reflexivity is obtained with a single predicate, we can revise (42) as (51).

In this sense, the predicate in sentences such as (52) is reflexive. In (52), *Pasha* is in the Spec of TP/VoiceP and *her* in Spec of DP, which is the complement of the single predicate *love*.

¹³ *Him* in *himself* is not a genitive form regardless of its status as a possessor. This has to do with the way features such as phi and case are assembled in syntax and particular spell-out rules. See Bai (2024, manuscript) for a detailed discussion.

¹⁴ See Bai (2024, manuscript) for more details of reflexivity characterized in this way.

R&R does not treat such cases as reflexive since the bindee must be an internal argument and *her* in (52), for example, is not an internal argument. R&R’s view is incorrect, however. As evidenced by the corresponding Mongolian sentence in (53), RX, being the hallmark of reflexivity, is present.

- (53) Pasha (öör-in) muur- ϕ -aa hairla-na.
Baatar-NOM self-GEN cat-ACC-RX love-NPST
‘Pasha loves her (own) cat.’

Importantly, the anaphor *öör* may be absent from the surface structure, indicating that anaphors are not obligatory reflexive markers cross-linguistically. This points to the possibility that in (52), there is a covert anaphor within the reflexive-bound DP, *her cat*. Bai et al (2025; manuscript) present reasonings for this, arguing that the adjectival *own* in fact functions as a phi-less anaphor, which can be combined to possessive personal pronouns to function as possessive anaphors.

What is important is a reflexive-predicate is not necessarily reflexive-marked in languages such as English, contrary to R&R’s claims, and this is the case with bound possessive pronouns as in (52). The absence of an anaphor in possessive DPs containing a bound pronoun cannot be used as evidence that reflexivity is not involved in sentences such as (52) as well as (53). R&R contend that morphological absence of an anaphor in such DPs indicates that reflexivity is absent in them. However, this turns out to be incorrect given that absence of evidence is not evidence of absence. On the other hand, cross-linguistically, anaphors, for example, Chinese *ziji* (54), Japanese *zisin* (55), and Mongolian *öör*, can easily be recovered in such DPs, without affecting the reflexive interpretation.

- (54) Pasha_i xihuan ta_i (ziji)-de mao.
Pasha love she (self)-GEN cat
‘Pasha loves her (own) cat.’
(55) Pasha_i kanozyo_i (zisin)-no neko-o ai-suru.
Pasha she (self)-GEN cat-ACC love-NPST
‘Pasha loves her (own) cat.’

All this said, reflexivity undoubtedly arises between two Specs. This leads to the unified analysis of reflexive binding in nominals (DPs) and clausal structures (TPs), as will be discussed in section 6.

5. Internal organization of anaphors and possessive pronouns

In this section, we elaborate on the internal structure of anaphors on the basis of the discussion in section 4. The core idea is that anaphors as well as possessive pronouns are formed in a portmanteau fashion. Technically, (an element with) [phi] sits in Spec, [poss] on D, and [n] on Comp, in the possessive DP structure, as represented below.

- (56) [_{DP} [phi] [_D [poss] [_N [n]]]]

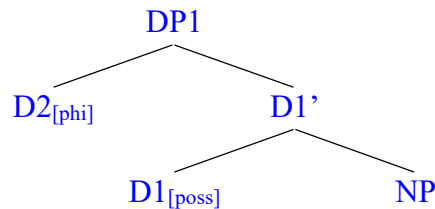
To elaborate on this, we need to look at first how possessive pronouns are derived, taking English as an example.

Previous studies on English (Abney 1987; Corver 1992; Chomsky 1995b; Munn 1995) suggest that possessors are merged in the specifier of DP, whose head D may be realized as *’s*. Hudson (2003) and Deal (2006), however, argue that English possessive pronouns are

portmanteau morphemes, which arise as result of morphological mergers, spelling out multiple syntactic nodes. Davis (2023) argues that English possessive pronouns are lexical realizations of phi-features ([phi]) and a possessive feature ([poss]), which enter the derivation separately but undergo clustering, ending up in a single D head. Davis (2023) assumes that either fusion (Halle and Marantz 1993; Embick and Marantz 2008) or spanning (Bye and Svenonius 2012; Merchant 2015; Haugen and Siddiqi 2016; Svenonius 2016) plus vocabulary insertion may lead to creation of possessive pronouns. The portmanteau formation of possessive pronouns prevents the D head from being realized independently, which, Davis (2023) argues, is the reason for why English possessives do not allow extraction out of them. Davis (2023) also assumes with Postal (1969), Abney (1987), Baltin (2012) and Stanton (2016) that (English) pronouns, lacking a lexical core (Wiltschko 1998), are non-projecting determiners.

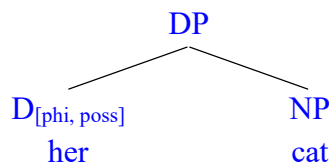
Specifically, possessive pronouns spell out the fused outcome of D and the material in its Spec. Based on a bare phrase structure theory of labeling (Chomsky 1995a, b), in which non-projecting heads are equivalent to phrases, Davis (2023) assumes that a bare determiner D2 occupies Spec of D1. Applying this analysis to *her cat*, we obtain (57).

(57) Initial structure of *her cat*



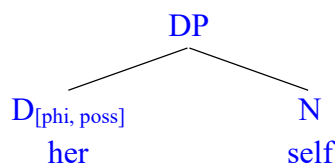
After (57) is built, D2 and D1 undergo fusion,¹⁵ thereby creating a single terminal node, where the features on D1 and D2 are clustered. The vocabulary insertion (VI) rule then applies, with the morpheme *her* is chosen for D, as in (58).

(58) Fusion of possessive D and its Spec



Note, however, that the above structure represents the non-anaphoric use of *her*, as in *Pasha_i loves her_j cat*, in which no reflexivity is obtained. Regarding the anaphor *herself*, we obtain the following structure, in which D_[phi] and D_[poss] undergo fusion, which is followed by spanning. Ultimately, *herself* spells out two terminal nodes, namely, D_[phi, poss] and N.

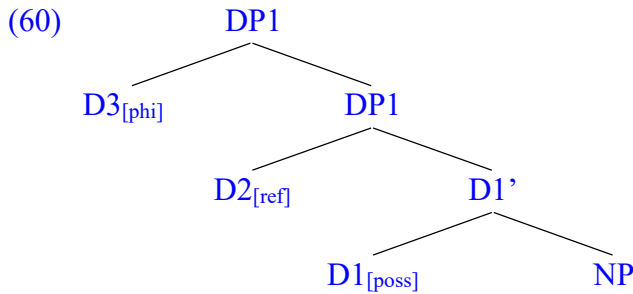
(59) Fusion of possessive D and its Spec



But what about *her* in its anaphoric use, as in *Pasha_i loves her_i cat* and possessive anaphors in many languages? It has been claimed that English does not have a possessive anaphor, also

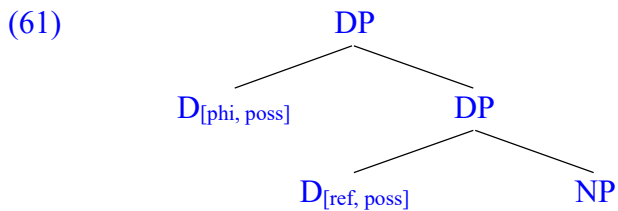
¹⁵ *Fusion* here, a term of Distributed Morphology (Halle and Marantz 1993; Embick and Marantz 2008), refers to a syntactic operation that gets two (or more) nodes united into one before the application of the morphological operation VI.

known as reflexive possessive (Truswell 2014). However, many other languages have one, e.g., Chinese *zi-ji de* ‘one’s own’ and Japanese kare *zi-sin no* ‘his own’. Let us first derive *her* in its anaphoric use. The two instances of *her*, I argue, are formed in the same pattern but differ in their feature organization. If possessive pronouns are realizations of D with [phi] and [poss], it is reasonable to say that possessive anaphors (reflexive possessive pronouns) are also realizations of D with appropriate features, among which there are [phi] and [poss]. The third feature is supposed to be (unvalued) [ref], for reflexivity. I assume that there is another bare D that bears [ref], which is stored in the lexicon, along with [phi] and [poss]; they are not added during the process of derivation in syntax, complying with Inclusiveness Condition (Chomsky 1995b). Because D with [poss] should be the head of a possessive phrase, D with [ref] as well as D with [phi] sits in Spec of D with [poss] in the initial structure, as shown below. The features [phi], [ref] and [poss] enter into the derivation separately. That is, anaphors, which encode [ref] and [poss] as well, are often deficient in phi-features (Reuland 2018: 2) and not all anaphors have [poss].



Note that four kinds of sets come out of these three features, namely, [phi, ref], [phi, poss], [ref, poss] and [phi, ref, poss], as inputs for clustering. However, clustering of features as well as fusion of heads is not arbitrary. It should be the case that head-head (lateral), spec-head (upward) or head-comp (downward), but not spec-spec, comp-comp or spec-comp, is subject to fusion. Given that D3 and D2 are both specifiers, they do not undergo fusion. That is, [phi] and [ref] do not undergo clustering, with the presence of [poss]. When [poss] is present with other features, it is subject to feature-clustering. D_[phi] and D_[ref] may undergo fusion if one of them is the head of the relevant DP, with D_[poss] absent. If D_[poss] is absent, no NP is selected by D_[ref]. This means that non-possessive pronouns spell out a D head that lacks a lexical core (Postal 1969; Abney 1987; Wiltschko 1998; Baltin 2012; Stanton 2016).

Regarding the above structure, fusion applies twice. That is, the ultimate structure looks like (85), which is a layered DP structure. D_[phi, poss] is spelled out by *her*. What about D_[ref, poss]? I argue that it is spelled out by *own*. This means that *own* is a possessive anaphor just like, for example, Mongolian *öör-in*. This said, English has a reflexive possessive pronoun, contrary to the previous claim.



Note that *self's* does not spell out D_[ref, poss], because *self* is chosen for N, as in (61). This would

give one an impression of the lack of possessive anaphor in English.¹⁶ *Own*, however, may be absent at PF, resulting in *her* in the anaphoric use, as in (62).

(62) Pasha_i loves her_i (own) cat.¹⁷

In Mongolian, $D_{[\text{phi}, \text{poss}]}$ and $D_{[\text{phi}, \text{poss}]}$ are spelled out by *tüün-ne* ‘her/his’ and *öör-in* ‘self’s/own’, respectively. However, *tüün-ne* is mostly absent at PF, arguably because of a language-specific morpho-phonological principle, which favors a more economical spell out. When it produces a distinctive interpretation such as an intensive interpretation, *tüün-ne* is most likely overt. If a distinctive interpretation is not obtained, the presence of *tüün-ne* lowers the acceptability of the sentence. Similar facts are observed in languages such as Chinese, in which *ta* ‘he/she’ in *ta zi-ji* or *ta zi-ji de* ‘her/his own’ may be absent if there is no semantic contribution it can make.

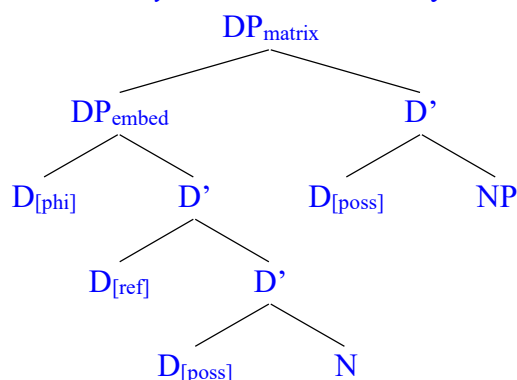
(63) #Baatar (tüün-ne) öör-in nom- ϕ -oo mart-san.
 Baatar-NOM 3SG-GEN self-GEN book-ACC-RX forget-PST
 ‘Baatar forgot his own book.’

Now we discuss the structure of the whole possessive DP structure containing a possessive anaphor. We clarify this using Mongolian complex anaphor *öör-in bey-in*.

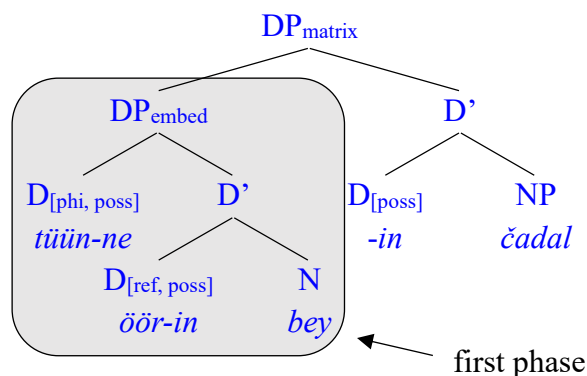
(64) Baatar öör-in bey-in-ee čadal-ig šalg-san.
 Baatar-NOM self-GEN body-GEN-RX strength-ACC examine-PST
 ‘Baatar examined his own strength.’

Two possessive D heads are involved in this structure. This means that the DP headed by the embedded $D_{[\text{poss}]}$ is located in Spec of the matrix $D_{[\text{poss}]}$.

(65) Initial structure of *öör-in bey-in čadal* ‘self’s body’s strength’



(66) Ultimate structure; lexical insertion applied



¹⁶ Bai et al. (2025) reasons out that *own* in English is in fact a reflexive possessive pronoun (possessive anaphor).

¹⁷ This means that such sentences are subject to Principle A, not Principle B, with no violation of it.

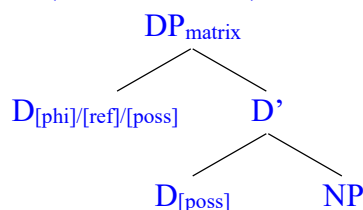
Due to economy principles of morpho-phonology such as (67), a spell-out with the smallest number of syllables, without affecting the possessive reflexive semantics, is favored.

(67) Use as few morphemes as possible, where other principles, if any, leave the choice open.

Consequently, either *öör-in* is the most likely ultimate realization of the embedded DP. *Bey-in* is also a likely option. However, due to the degree of lexicalization of *bey* as a body anaphor and the retention of the prototypical meaning “body” to some extent, *bey-in* is less likely to be used in possessive DPs expressing an alienable possession as in *my book*. With inalienable possession as in *my strength*, *bey-in* is more likely to be used, as in *bey-in čadal*, which literally means “body’s strength”.

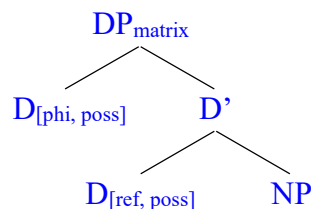
Importantly, since the embedded DP (DP_{embed}) as a whole can act as a simple element with [phi] or [ref], as represented below.

(68) DP_{embed} as D (initial structure)



This means that DP_{embed} as a possessive DP in fact represents the first phase formation of a bare phrase (or a bare D) located in Spec of a matrix D_[poss].¹⁸ This in turn means that when [phi], [ref], and [poss] are selected from the lexicon into Spec of the D_[poss], what would otherwise spell out the embedded DP (DP_{embed}) in (66) as a whole spells out the D_[phi] or D_[ref] part of D_[phi, poss] or D_[ref, poss] in (69), which is the ultimate structure of (68).

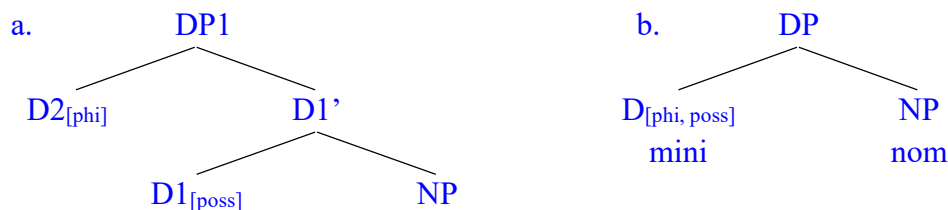
(69) DP_{embed} as D (ultimate structure)



The possessive feature [poss] on D in the first phase structure is thus squeezed out in the ultimate structure of the matrix DP. Therefore, DPs such as (*mini*) *öör-in nom* ‘self’s/own book’ have either of the structures in (66), which is the full structure containing the first phase derivation, or (69), which is the very ultimate structure.

Regarding non-reflexive possessive DPs, we obtain the structure in (70).

(70) Formation of non-reflexive possessive DPs (*mini nom* ‘my book’)



¹⁸ The term “first phase” here, borrowed from Ramchand (2008), refers to a property that an element has a word-internal phrasal structure, which is unfolded to interplay with or have an impact on the syntactic properties of that word. Informally, the word-internal phrasal structure can be called “first phase structure”.

After (70a) is built and before the VI is applied, D2 and D1 undergo fusion, thereby creating a single node, with [phi] and [poss] clustered on it, as in (70b). Then, the VI applies and *mini* is inserted into D_[phi, poss].

Having elaborated on the internal organization of anaphors, we move on to how binding takes place during the process of their portmanteau derivation.

6. Explaining reflexive binding

We first derive the distinction of predicate-internal binding and predicate-external binding and discuss its implications for some of current debates on binding in section 6.1, and implement binding in section 6.2.

6.1. Predicate-internal binding vs. Predicate-external binding

Reuland (2001, 2011, 2020a, 2020b) among many others presents a line of reasoning in which binding is Agree, which is an operation for DP subjects to value phi-features on anaphors. Reuland assumes that phi-features of anaphors start from Spec of TP in a binding context, where Agree requires tense as well as Case to interact with it. This is a different approach than Kratzer (2009), who argues that phi-features of anaphors start out from the specifier of vP. For Kratzer (2009), binding is not feature-valuation, but rather feature-transmission, which is not Agree per se. The essential difference between feature-valuation and feature-transmission is that for the former, unvalued phi-features are encoded by anaphors from the beginning, while for the latter, anaphors come to have phi-features after they receive them from a local subject. Rooryck and Vanden Wyngaerd (2011) also assume that phi-features of anaphors originate from a local subject in Spec of vP. Rooryck and Vanden Wyngaerd (2011) argue that phi-features are specified for anaphors via Agree, which takes place between anaphors and local subjects merged in Spec of vP. Rooryck and Vanden Wyngaerd (2011), however, assume that anaphors are not bindees but binders, whereas local subjects are bindees, not binders. Several issues thus have been controversial: What mechanism is binding? How is it achieved? Where (vP or TP) is it completed? What defines a binding domain (vP or TP)? In what follows, I show that these controversies arise where binding is undistinguished between predicate-internal and predicate-external.

Binding must be distinguished between internal and external. In a nutshell, internal binding is completed within the predicate phrase devoid of any tense-aspectual inflection and external binding is completed with help by tense-aspectual inflection.

(71) Internal binding: [TP XP_i [^{VoiceP} t_i [^{VP} ... YP_i]]]

(72) External binding: [TP XP_i [^{VoiceP} t_i [^{VP} ... YP_i]]]

(73) Predicate-internal binding vs. predicate-external binding

Predicate-internal binding	Predicate-external binding
binder located within the predicate	binder located outside the predicate
not intertwined with tense/inflection	intertwined with tense/inflection
vP as binding domain	TP as binding domain
not local	local

may or may not be reflexive-marked	reflexive-marked
------------------------------------	------------------

Crucial is dynamic positions of the subject, which originates within the predicate but may appear outside it. Given that binding as Agree involves phi-features on a subject, it is a dynamic, streamlined, and derivational process. With tense/inflection assigned to the predicate and the subject promoted to Spec of TP, binding comes to be reassessed. Reassessed binding is on par with chain-based binding advocated by Reuland (2011) in that both involve Agree plus Move. They, however, differ in that the former starts as internal binding, which is already completed within vP, but the latter is not completed until the binder occurs in the inflectional domain. The independence of internal binding on verbal inflection as well as subjective case is evidenced by the fact that a predicate-internal subject (of an uninflected predicate) can successfully bind an anaphor, as in Mongolian as well as Japanese. In Mongolian, for example, the agent (*Dorž* in (74)), being the subject of the embedded VoiceP, can bind the anaphor, in contrast with the case such as (75), in which the causer *Baatar*, occupying Spec of the inflected predicate phrase, binds the anaphor. Importantly, the two instances of binding employ different markers. Only in the latter is RX employed.¹⁹

(74) *Baatar*_i [VoiceP *t*_i [VoiceP *Dorž*_j-oor *öör*_{i/*j}-ig-ni *šüümjil*] -üül] -sen.
Baatar-NOM *Dorž*-INS self-ACC-PSS[3] criticize-CS-PST
 ‘Baatar made Dorž criticize himself (= Dorž).’

(75) *Baatar*_i [VoiceP *t*_i [VoiceP *Dorž*_j-oor *öör*_{i/*j}-ig-**öö** *šüümjil*] -üül] -sen.
Baatar-NOM *Dorž*-INS self-ACC-RX criticize-CS-PST
 ‘Baatar made Dorž criticize him (= Baatar).’

In (74) and (75), the agent is located within the predicate, which we argue is VoiceP, following Bai (2026), and binds the possessive anaphor. Not being the hierarchically highest argument in the voice/predicate domain, the agent is not promoted to the Spec of TP, but the causer, which is the highest argument, is. That is, for an in-situ subject, which is not a highest argument, binding remains as internal binding, and for a derived subject, which is a highest argument, binding starts as internal but becomes external binding.²⁰

The re-assessment is re-marked where the language, for example, Mongolian, provides distinct morphology than the anaphor available as the marker. Where no distinct morphology is available, the anaphor remains as the marker. Cross-linguistically, internal binding of possessors may or may not be anaphoric-marked, as in (52–55).

Languages such as English, Chinese, and Japanese do not necessarily distinctively mark external and internal bindings due to the lack of relevant morphology in the morphological inventory in them; that is, re-assessment of binding within TP is left unmarked in these languages. In contrast, Mongolian marks both internal binding and external binding of both possessors and objects.

One might predict that external binding should be anaphoric-marked in order to be

¹⁹ The fact that RX, the reflexivizer, is present only for external binding suffices to show that external, not internal, binding is the canonical type of reflexive binding. That is, verbal inflection plays an important role in deriving external binding, the canonical reflexive binding, from internal binding.

²⁰ Borjigin and Bai (2025) justified that Reflexive-Possessive Principle in Mongolian is a type of Binding Principle A of the standard binding theory, which further demonstrates that external binding indeed exists as an independent type of binding, which is composed of Agree and Move and overtly marked by RX.

(77) [TP [VoiceP sbj_[phi] [DP ... D_[phi, poss] ... D_[ref, poss] ... [NP ...] ...] ...]
↑ ↑
mini öör-in

D_[ref, poss], however, is irrelevant to the implementation of binding; it is ([phi] in) [phi, poss] that is relevant. After valued, [phi, poss] is assigned the morpheme *mini*, for example. With binding established, [ref] in [ref, poss] gets evoked and a **reflexive** interpretation is produced.

The proposed analysis successfully accounts for binding of accusative and genitive anaphors as well as binding into nominals and inflected clauses, with or without *gež*, in a unified way. As summarized in (12) and (13), a matrix subject can bind the accusative subject of object clauses with *gež* (78) and the genitive subject of object clauses without *gež* (79). Note that such object clauses are all inflected. With *gež*, they are CP, and without it, they are TP.

(78) Baatar öör-ig-öö buruud-san gež med-sen.
Baatar-NOM self-ACC-RX go wrong-PST COMP know-PST
‘Baatar realized that he was wrong.’ (=7)

(79) Baatar (öör-in) buruud-san-aa meder-sen.
Baatar-NOM self-GEN go wrong-PST-RX admit-PST
‘Baatar admitted that he was wrong.’ (=9)

We need to first determine the position of the genitive and accusative subjects. Fong (2019) observed that the accusative subject of *gež* clauses is raised to Spec of CP, thereby becoming accessible from the matrix v. Thus, *öör* of *öör-ig-öö* in (78) is within the minimal matrix phase as the binding domain and is bound by the matrix subject in Spec of vP, as illustrated below.

(80) [TP [vP Baatar_i [CP öör_i-ig-öö C [TP t_i [vP t_i buruud] -san] gež] med] -sen]

This is exactly what is predicted in Bošković’s (2016) formulation.

(81) [A]n anaphor can be bound outside of its own minimal phase XP only if it is located at the edge of the phase (the anaphor then does not really ‘belong’ to phase XP; rather, it belongs to a higher phase). (Bošković 2016: 14)

Regarding the genitive anaphor *öör-in* as the subject of complement clauses (79), this paper argues that it is located on Spec of DP that contains a clausal structure. Specifically, TP is selected by D_[poss] with the *HOLD* semantics. This is plausible given that TPs are subject to nominalization (Kornfilt and Whitman 2011; 2012). In this sense, the genitive subject is some kind of possessor that holds the event as a possessum under its control. Gong (2023) reasoned out that TP nominalization takes place in Mongolian. All this said, this paper argues that the genitive case marker *-in* in *öör-in* is the spell-out of the possessive D head, whose Spec is occupied by *öör*, as illustrated below.

(82) [TP [vP Baatar_i [DP öör_i D(=in) [TP [vP t_i buruud] -san-aa]] meder] -sen]

To elaborate, a bare D with an unvalued [phi] and [ref] is introduced as an external argument by v, and subsequently moves to Spec of DP. Since *öör-in*, located in Spec of DP, is outside of the minimal phase DP (D’ traditionally), it is accessible from the matrix clause. This makes it possible that [phi] on *öör* is valued by that on the matrix subject via Agree/binding, thereby evoking [ref]. After fusion applies, VI chooses the morpheme *öör-in* for D_[ref, poss]. D_[phi, poss] is realized as the third person singular possessive pronoun *tüün-ne*, which, however, is not absent at PF and even prevented due to economy principles such as (67). Interestingly, such principles even allow D_[ref, poss] to be absent, as exemplified below.

(83) Baatar buruud-san-aa meder-sen.

Baatar-NOM go wrong-PST-RX admit-PST

‘Baatar admitted that he was wrong.’

Importantly, binding holds and reflexivity is obtained even when the (genitive) anaphor is not overt. This is why RX is present. Note that the presence of RX on *öör-in* is disfavored, arguably because the structure in (66), in which an embedded possessive structure is contained, is less economic and less preferred compared with the structure in (69), which does not contain an embedded possessive structure. What occupies Spec of DP is thus a simple *öör*. If no possessive structure is contained, the semantic predicate *HOLD* does not hold. This means that for *öör-in* therein is not a possessum that can host RX. RX is thus present elsewhere, thus deriving EMC in (14).

The same holds true in the case of relative clauses, in which the relative subject is the genitive anaphor *öör-in*, exemplified below.

(84) Baatar (öör-in) hel-sen üg- ϕ -ee mart-san.
 Baatar-NOM self-GEN-RX say-PST word-ACC-RX forget-PST
 ‘Baatar forgot the words he said.’ (= (11))

As shown in (85), the relative clause is located in Spec of NP, and a bare D with unvalued [phi] and [ref], which is base-generated in Spec of vP, moves out from the clause (TP/RC) to Spec of DP, headed by D_[poss]. Fusion applies so that D_[ref, poss] and D_[phi, poss] are derived. Since [phi] is now within the same domain as the matrix subject, it gets the value of [phi] on the matrix subject and [ref] is evoked. VI applies and *öör-in* spells out D_[ref, poss], as is the case with (82).

(85) [_{TP} [_{vP} Baatar_i [_{DP} öör_i D [_{NP} [_{TP(=RC)} [_{vP} t_i hel] -sen] üg-ee] mart] -san]

It is important to note here that locality in cross-clausal binding is achieved by raising the embedded subject to the edge of the maximal CP phase for *gež* clauses and by nominalizing clauses without *gež* into possessive DPs and locating the anaphoric element in their edges. Such cross-clausal binding and within-clausal binding share a common structure mimicking a possessive DP, where reflexive binding is a linkage between specifiers, not simply between a specifier and a complement, in a local domain.

7. Conclusion

Mongolian displays several important binding properties that makes it differ from other languages. **First**, the hallmark of binding in Mongolian is not an anaphor (*öör*) but rather a suffix (*-aa*); a bound possessive anaphor can be absent at PF but the suffix in question cannot in a local binding domain, where locality is determined by verbal inflection. Second, local binding is obligatorily RX-marked and optionally anaphoric-marked, whereas remote binding, which is not reflexive, is just anaphoric-marked. **Third**, binding of an embedded subject by the matrix subject takes place only for inflected object clauses and object relative clauses. **Fourth**, local binding is canonical reflexive binding, whereas remote binding is not.

At least six important implications arise from the Mongolian reflexive binding strategies and the proposed analysis of them, suggesting more general cognitive properties concerning binding, reflexivity, and anaphors. **First**, binding is derivational, not representational; binding domain is contextual, not fixed; the position of binders is variable, not uniform; crucially, the position of subject as binders entails the morpho-syntactic properties of binding; crucially, the inflectional properties of the predicate entailing Move of subjects derives predicate-external binding from predicate-internal binding. **Second**, binding is feature-valuation, not feature-

transmission, where the values of phi-features on a nominal occupying Spec of vP are copied to a c-commanded anaphor, instantiating Agree. **Third**, the size of clauses required for locality is universally the same but the ways of achieving locality vary between languages, where the case-marking system and the morpho-syntactic properties related to the tense-inflection system (or TP-FinP-CP) and clause-nominalization in individual languages play important roles. **Fourth**, anaphors are formed from a possessive feature, phi-features, and a **reflexivity** feature, which enter the derivation of a possessive DP separately but come to be clustered, followed by vocabulary insertion; a possessive feature is always a head, with other features being its specifiers; anaphors thus DPs with a possessive, not capsule-like, structure; a first phase derivation and language-specific spell-out rules play important roles in forming and interpreting anaphors, especially body(part) anaphors. **Fifth**, reflexivity is an identity relation between specifiers, where the binder occupies a subject, internal or derived, position, and the bindee occupies the (highest) specifier in the complement with respect to the same predicate; that is, a predicate is reflexive where its co-arguments co-refer by means of binding into the specifier of the c-commanded argument. **Sixth**, there is a deeper structural link not only between anaphors and possessives but also between possessive nominals and nominalized clauses.

The primary contribution of this paper lies in the division of predicate-internal binding and predicate-external binding and technical elaborations of how binding steps into the internal structure of anaphors, which are the key to resolve the debated issues about what mechanism is exactly. We have tried to demonstrate that the Mongolian facts about reflexive binding can serve as morphological evidence for or against different current theories of binding. We have offered a novel argument in favor of the binding-as-phi-agreement (binding-as-feature-valuation) approach (Reuland and Everaert 2000; Reuland 2001, 2011, 2020a, 2020b; Reuland and Zubkov 2022), rejecting the binding-as-feature-transmission approach (Kratzer 2009). The proposed analysis lends support to the claim that a binding theory applies in the narrow syntax, not at LF (Hicks 2008, 2009; Reuland 2011).

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